

Appendix

System Details

Incidents are categorized into one of 5 types: clinical (which do or could impact the clinical outcome of the radiation therapy intervention), operational, occupational, environmental and security or other. Four levels of severity with respect to radiation dose and volume deviation are defined: critical, major, serious and minor, details given in Table A1. (The definition of an incident in radiation treatment differs from that in other areas of medicine where the concern may be immediate harm such as patient falls, medication errors, etc.. Due to the latent period of radiation induced effects, harm in radiation treatment is generally not immediate and therefore cannot be based on a level of visible injury. Rather the harm results from a deviation from intended dose and/or its location within the patient and manifests as excess morbidity or sub-optimal therapeutic benefit.)

The system defines domains in which an incident could occur or be discovered, viz., dose prescription, booking/scheduling, treatment preparation and treatment delivery. Incidents discovered during routine quality control performed within the domain in which they occurred reflect normal system functioning and are not entered into the system. However, those events which propagate out of the domain of occurrence and thus demonstrate a failure of the quality control program are classified as incidents. Following investigation, the basic cause(s) is determined following the standard method of asking a series of “why” questions. The classification of basic causes follows a scheme with 9 categories: 3 relating to job factors, 3 to systematic or management factors and 3 to personal and natural factors. Based on the immediate causes of the incident, as well as the underlying cause(s) identified through the causal analysis, recommendations for corrective actions to prevent a recurrence of the incident are developed. The final step in the system is the learning component, accomplished in several ways including the presentation of summary data to program staff at all levels from leadership to front-line therapists, email alerts and the presentation of selected incidents with a high learning potential.

The effectiveness of the incident learning system was assessed from the ratio of the number of potential and minor incidents to the number of serious, major and critical incidents, termed the reporting index. The larger the ratio, the more the minor or potential incidents are reported and the better the reporting performance. In an effective incident learning system, paradoxically the overall total number of incidents reported is expected to increase reflecting an increasing participation rate. At the same time the ratio of the number of potential and minor incidents to the number of serious, major and critical incidents should also increase indicating a culture of tuning into the weak signals within the system and responding, thus improving patient safety.

Implementation Strategy

The first and most important step in the implementation of the incident learning system was to secure a commitment from program management to support and understand the inherent value of an incident learning system to foster a culture where staff feel safe to report incidents. This includes psychological safety (providing support to those involved in incidents through services such as Employee Assistance Programs), and professional safety, where staff know their job security will not be jeopardized.

A necessary condition for the success of an effective incident learning system is the existence of a just environment in which every staff member is encouraged to freely report all incidents without repercussion. A further essential feature of the system is that it requires an investigation of sufficient depth to discover the basic causes of an incident according to a predetermined categorization. Without such an approach, the identification of meaningful corrective actions is compromised. An added benefit of such a comprehensive approach is that it enables the identification of system problems or basic causes that could precipitate a range of different incidents. If the basic causes are addressed, it can be expected that overall system safety is enhanced rather than just a particular weakness associated with a particular incident.

A multidisciplinary implementation team was formed to generate documentation, policies and procedures and educational material. Documentation and educational materials from the Alberta Heritage Foundation for Medical Research were customised for our use including a comprehensive reference guide, system specific policies and procedures, a tracking spreadsheet, a committee structure, educational presentations and hand-out material for each professional group.

Considerable effort was made to design a report form that was easily completed in less than 5 minutes. Forms were printed and placed in appropriate locations throughout our facility. An implementation date of 1 September 2006 was identified. During July and August 2006, information sessions were held to educate all staff on the use of the system as the first step in establishing a non-punitive, just environment.

Ongoing Management of the System

The system is managed by a multidisciplinary team which includes the Chiefs of Radiation Oncology, Radiation Therapy and Medical Physics, and videoconferencing is used to engage leadership at multiple sites. This team meets weekly to direct the investigation process, discuss new incidents and determine corrective actions and follow up, as required. With an average of approximately 10 incidents reported per week, addressing the reports on a weekly basis, although resource intensive, is a key strategy in ensuring prompt response and feedback. All incidents are investigated to determine basic causes and identify corrective actions to prevent reoccurrence.

Although each incident is investigated, we have found that the investigation of the majority of incidents may be addressed individually by a small number of key supervisors, depending on the nature of the incident. A full team investigation (root cause analysis) is used for the more serious incidents and for all cases where determination of basic cause and identification of the most appropriate learning component requires more rigorous assessment. Most incidents are investigated within days of the report being filed ensuring

prompt feedback through weekly discussion of the details. Appropriate corrective actions and a strategy for learning (and communication) are assigned to a team member, or members.

Additional Results

Table A2 gives a summary of actual clinical incidents and Table A3 a listing of the major program changes during the reporting period.

Table A1: Categorisation of severity for clinical incidents

Severity	Examples
Critical	• Radiation dose or medication error causing death or disability • Dose variation from prescribed total dose of >25% • Completely incorrect volume or wrong patient • Tolerance dose exceeded to critical structure
Major	• Radiation dose or medication error causing side effects requiring major treatment and intervention or hospitalisation • Dose variation from prescribed total dose of 10 – 25% • Incorrect beam parameters or shielding for > 10% of fractions • Set up variation that may impact normal tissue effects (i.e. heart, lung, eyes, kidney) • Breaches of protocol impacting patient
Potential Major - A near miss that could have been a major incident	
Serious	• Radiation dose or medication error causing side effects requiring minor treatment or ongoing monitoring and assessment • Dose variation from prescribed total dose of 5 – 10% • Set up variation >1cm, no critical structures included.
Potential Serious - A near miss that could have been a serious incident	
Minor	• Dose variation from prescribed total dose of <5% • A near miss or unsafe condition which could potentially cause a treatment error • Patient complaint

Table A2: Summary of actual clinical incidents

Category	2007	2008	2009	2010	2011
Geographic miss	10	8	5	3	1
Incorrect shifts from setup point	5	4	1	6	7
Incorrect patient treated	2	-	-	-	-
Incorrect volume	1	-	1	-	1
Shielding incorrect	15	4	-	1	4
Incorrect treatment parameter(s)	12	8	3	-	-
Patient positioning error	2	1	3	2	1
CT errors	8	11	4	6	4
Imaging errors	10	4	5	5	2
Bolus issues	14	3	3	2	10
Incorrect accessories	2	2	2	2	2
Incorrect dose/calculation error	5	4	4	2	1
Scheduling errors	2	3	3	1	6
Miscellaneous	-	3	2	-	-
Total	87	54	36	30	39

Table A3: Major program changes during 2007-2011

Technology	o 4 new treatment planning software systems
	o 3 new structure contouring software systems
	o 2 new types of treatment delivery units
	o 2 different image-guidance systems
	o 1 add-on beam shaping device

Techniques	o Image guidance for target positioning for treatment delivery
	o Extra-cranial stereotactic treatment delivery for lung, liver and spine
	o Intensity modulated radiation therapy (IMRT) both fixed field and dynamic arc delivery
	o Intra-operative radioactive seed placement for interstitial brachytherapy

Geography	o Construction and commissioning of substantial new facilities
	o Relocation of brachytherapy operations
	o De-commissioning of facilities at one of the two original sites
	o Relocation of physics office space – twice!
